Environmental Health Risk Assessment

Graduate / Professional Course Design

Course Format

A complete 14-week course design for graduate and professional instruction, with measurable objectives, applied cases, aligned assessment, and a cumulative risk-assessment project.

Course Overview

Environmental health risk assessment turns incomplete evidence into a transparent account of what may cause harm, who or what may be exposed, how large the risk may be, and what remains uncertain. This graduate/professional course develops the full reasoning sequence: problem formulation, conceptual modeling, hazard identification, dose-response evaluation, exposure assessment, risk characterization, and communication for decision support.

The course treats risk assessment as a scientific input to decisions, not the decision itself. Students distinguish evidence from assumption, variability from uncertainty, regulatory requirements from professional guidance, and risk-assessment conclusions from the value, feasibility, equity, and policy judgments involved in risk management.

Intended Audience

Graduate students and professionals in environmental health, public health, occupational health, environmental science, environmental engineering, preventive medicine, or related fields. Learners should be comfortable interpreting scientific evidence and performing transparent calculations; no particular software platform is required.

Course Learning Objectives

By the end of the course, students will be able to:

1. Frame an environmental-health risk problem by specifying the decision context, population, stressor, scope, and analytic boundaries.
2. Construct and defend a conceptual model linking sources, pathways, routes, receptors, outcomes, and potential interventions.
3. Evaluate hazard and dose-response evidence for relevance, quality, limitations, bias, and appropriate use.
4. Design or critique an exposure assessment that identifies populations, routes, magnitude, frequency, duration, data needs, and model assumptions.

5. Integrate hazard, dose-response, and exposure evidence into a risk characterization that distinguishes variability, uncertainty, sensitivity, and data gaps.
6. Distinguish scientific risk assessment from risk-management choices and interpret requirements, guidance, and recommendations according to their authority and current status.
7. Conduct a transparent, reproducible analysis using documented sources, assumptions, calculations, and versioned work products.
8. Communicate a bounded, defensible risk conclusion and decision-support recommendation to technical and nontechnical audiences.

Teaching and Learning Approach

Preparation is deliberately manageable: short readings, source checks, concept questions, case background, data previews, or brief written reasoning. Class time is used for application - comparing evidence, drawing conceptual models, checking calculations, testing assumptions, discussing cases, and defending professional judgments.

Four original cases provide a cumulative practice environment: a PFAS drinking-water concern, workplace solvent use, extreme heat and population vulnerability, and a community vapor-intrusion concern. Participation can take the form of written reasoning, calculations, source evaluation, small-group analysis, peer critique, discussion, or a defended recommendation. The objective is to make reasoning visible, not to reward speaking frequency.

Course Materials and Readings

Readings are selected for the problem at hand and checked for authority, scope, and currency. Students use current official guidance, peer-reviewed evidence, transparent models, and small public or project-generated datasets. Each learner maintains an evidence ledger recording source, version or access date, claim supported, limitation, and whether the source is regulation, guidance, recommendation, data, or research evidence.

Assessment and Grading

Assessment	Weight	Evidence of learning
Preparation and short analyses	15%	Source review, framing, assumptions, readiness
Applied exercises	20%	Models, calculations, exposure scenarios, uncertainty
Case analysis memorandum	20%	Integrated reasoning and professional communication
Cumulative risk-assessment project	30%	Transparent assessment and decision support
Final briefing and defense	15%	Audience-centered communication and judgment
Total	**100%**	

Assessment emphasizes application rather than a high-stakes recall examination. Criteria include scientific reasoning, source quality, transparency, appropriate methods, treatment of uncertainty, distinction between assessment and management, and communication proportional to the evidence.

Major Risk Assessment Project

Students develop one bounded environmental-health risk assessment over the term. It may be presented as a technical report, decision memorandum, risk assessment, or professional briefing, but it must include:

1. a decision-centered problem statement and scope;
2. a conceptual model of sources, pathways, routes, receptors, outcomes, and interventions;
3. a documented evidence search and authority map;
4. hazard and dose-response evaluation appropriate to the question;
5. an exposure assessment with explicit variables, time frame, and assumptions;
6. a risk characterization addressing variability, uncertainty, sensitivity, and data gaps;
7. a boundary between scientific assessment and risk-management considerations;
8. a reproducibility package containing sources, methods, calculations, provenance, and version notes;
9. a bounded recommendation explaining what additional evidence could change the conclusion; and
10. a final briefing for a specified professional audience.

Projects use public, de-identified, or synthetic information. They do not authorize human-participant recruitment, private-record access, or field sampling requiring institutional approval.

Weekly Course Schedule

Wk	Topic	Preparation	In-class application	Deliverable
1	Risk, evidence, decisions	Official overview; classify claims	Diagnose a contested decision	Topic shortlist
2	Problem formulation and models	Case background; draft boundaries	Build and critique conceptual models	Problem statement
3	Hazard identification	Compare evidence types	Weight-of-evidence workshop	Evidence map
4	Dose-response	Review endpoint summaries	Interpret model and extrapolation limits	Evidence note
5	Exposure fundamentals	Preview variables and data	Scenario construction and calculation audit	Exposure plan
6	Environmental and occupational pathways	Review water and workplace cases	Compare routes, sampling, and controls	Exercise 1
7	Risk characterization	Read contrasting characterizations	Integrate evidence into a qualified conclusion	Outline

Wk	Topic	Preparation	In-class application	Deliverable
8	Variability and uncertainty	Complete assumptions table	Test influential assumptions	Sensitivity memo
9	Regulation and management	Classify sources by authority	Separate findings from management choices	Authority map
10	Drinking-water case	Review current PFAS sources	Draft bounded utility decision memo	Case memo
11	Workplace case	Review controls and exposure guidance	Design measurement and control strategy	Exercise 2
12	Vulnerability and emerging hazards	Review heat-health evidence	Prioritize action with incomplete evidence	Project draft
13	Risk communication	Draft technical and public summaries	Peer review for accuracy and proportionality	Revision
14	Presentation and synthesis	Prepare briefing and evidence ledger	Deliver, question, defend, and revise	Final project

Course Policies

Evidence and attribution. Every consequential claim, dataset, parameter, and regulatory statement must be traceable. Students distinguish quotation, paraphrase, analysis, and inference; cite version or access date for changeable content; and state when evidence is incomplete or conflicting.

Responsible use of artificial intelligence. AI tools may support brainstorming, editing, coding, or data workflows when allowed. Students remain responsible for source verification, methods, calculations, data protection, interpretation, conclusions, and disclosure of material AI contribution. Fabricated sources, undisclosed substitution for required reasoning, and uploading protected data to unauthorized systems are prohibited.

Reproducibility and transparency. Analyses must be reproducible from described inputs and steps. Assumptions, exclusions, transformations, relevant tool versions, and material judgment calls must be recorded. Precision must not imply more certainty than the evidence supports.

Professional responsibility. Students protect privacy, minimize sensitive data, represent affected populations respectfully, and avoid implying that classroom analysis is an official regulatory, clinical, or institutional determination.

Currentness and Professional Use

Risk frameworks are durable, but regulations, screening values, toxicity information, datasets, and agency recommendations change. Students and practitioners must verify controlling requirements and current official sources at the point of use. Proposed rules are not final requirements, recommendations are not regulations, and a screening value is not automatically a complete risk conclusion.

Selected References

- U.S. EPA. Risk Assessment. Accessed 2026-08-10.
- U.S. EPA. Human Health Risk Assessment. Accessed 2026-08-10.

- U.S. EPA. Guidelines for Human Exposure Assessment. Final agency guidelines, 2019; webpage accessed 2026-08-10.
- U.S. EPA. Risk Management. Accessed 2026-08-10.
- U.S. EPA. PFAS National Primary Drinking Water Regulation. Verify current requirements and proposals at use.
- NIOSH. Hierarchy of Controls.
- U.S. Global Change Research Program. Fifth National Climate Assessment, Chapter 15: Human Health.